Part 1. 1-2

Stemming comparison

The comparison between unstemmed and stemmed texts reveals clear differences in vocabulary structure and frequency distribution. Without stemming, morphological variants such as *love*, *loved*, and *loving* are treated as separate lexical items. As a result, the overall vocabulary size is larger and the frequency distribution is more dispersed across related word forms. After applying stemming, these morphological variants merge into a shared root form, which reduces the overall vocabulary size and increases the frequency of the resulting stems. Consequently, the frequency distribution becomes more concentrated, as multiple word forms are counted under a single stem. For example, *love*, *loved*, and *loving* are reduced to *love* by the Porter stemmer and to *lov* by the Lancaster stemmer.

A comparison between the Porter and Lancaster stemmers shows important methodological differences. The Porter stemmer is less aggressive and produces stems that remain recognizable. It applies more conservative reduction rules, which helps preserve readability and interpretability. For instance, *running* becomes *run* and *studies* becomes *studi*. Although the latter form is not a complete word, it remains clearly identifiable. In contrast, the Lancaster stemmer is more aggressive and frequently over-stems words. It often reduces words to shorter and less interpretable forms. For example, *running* becomes *run*, *studies* becomes *stud*, and *general* may be reduced to *gen*. Such reductions distorts meaning and reduces readability.

Overall, stemming significantly reduces lexical diversity by collapsing inflected and derived forms into shared roots. While both stemmers achieve this reduction, the Porter stemmer provides a more balanced approach, maintaining greater linguistic transparency. The Lancaster stemmer reduces vocabulary size more drastically but at the cost of interpretability. For literary texts such as these, the Porter stemmer appears to be the more appropriate choice.

Part 1. 3

POS frequencies

The POS tag frequencies across the English, Dutch, and German translations show both similarities and differences. Overall, the distribution of major word classes such as nouns and verbs is relatively similar across the three languages. This suggests that the narrative structure of the text remains consistent in translation.

However, some differences can be observed. German shows a higher proportion of nouns compared to English and Dutch. This suggests that German uses more noun-based constructions or compound nouns. Dutch generally appears to fall between English and German in this respect.

English shows a slightly higher frequency of auxiliary verbs (AUX). This is likely because English often uses auxiliary verbs to form tenses and aspects (for example, “was going” or “has done”). German may rely less on auxiliary constructions in comparison.

English also shows a relatively high number of determiners (DET), such as “the” and “a,” which are used very frequently in English. Differences in article usage across the languages may explain these variations.

Pronoun usage also differs slightly. English appears to use pronouns more frequently, possibly because they are always required in English sentences.

Overall, the findings suggest that while the translations preserve the same story structure, grammatical differences between the three languages are reflected in the POS tag distributions. These differences likely result from structural characteristics of each language rather than from translation choices.

Part 2

Manual NER

Sentence_1 = "Missy was sitting on a chair in a house, maps and papers spread around"

Missy = PERSON

sentence_2 = "River hadn't gotten lots of sleep since he visited"

River = PERSON

sentence_3 = "I was going to take you to so many places. Barcelona."

Bercelona = GPE

sentence_4 = "He had Rose and lost her. Now Martha is here to fill that gap in his life."

Rose = PERSON

Martha = PERSON

sentence_5 = "If the Doctor was there, you'd tell her she is an idiot."

Doctor = PERSON

SpaCy NER

sentence_1 = "Missy was sitting on a chair in a house, maps and papers spread around"

Missy = PERSON

sentence_2 = "River hadn't gotten lots of sleep since he visited"

NO ENTITY PREDICTION

sentence_3 = "I was going to take you to so many places. Barcelona."

Barcelona = GPE

sentence_4 = "He had Rose and lost her. Now Martha is here to fill that gap in his life."

Rose = PERSON

Martha = PERSON

sentence_5 = "If the Doctor was there, you'd tell her she is an idiot."

NO ENTITY PREDICTION

The spaCy NER system performs reasonably well in identifying standard personal names such as *Rose* and Martha. However, it struggles with fictional entities such as *Doctor*, which are central to the narrative but not consistently recognized as named entities. Overall, performance is moderate. The model performs better on conventional real-world names but shows limitations when applied to specific science fiction texts. This suggests that spaCy's pretrained model is optimized for general English and may require adaptation for fictional universes.

While annotating with spaCy, I came across an example that demonstrates pretty well the case. In the sentence *"River hadn't gotten lots of sleep since he visited,"* the model fails to assign any entity label to "River," despite it being a clear reference to a character. This indicates that the model does not consistently recognize "River" as a PERSON entity. Since "River" is also a common English noun referring to a geographical feature, the model likely treats it as ambiguous, therefore it does not recognize it, resulting in no entity prediction.

I realized this problem when I replaced the name "Rose" with "River" in sentence 4. There, spaCy identifies "Martha" as a PERSON entity but labels "River" as LOC (location). This misclassification can be explained by training data bias. In general-domain corpora, the word "River" frequently appears in geographical expressions (e.g., "River Thames"), which strengthens its statistical association with location entities. When faced with contextual ambiguity, the model appears to default to this learned pattern, resulting in incorrect labeling.

These examples illustrate the context-sensitive nature of pretrained NER systems. Entity recognition decisions are influenced by surrounding syntactic structure and patterns learned from the training data. It appears that spaCy is trained in standard English, therefore it struggles with domain-specific content such as science fiction. The case of fictional names overlapping with common nouns stand as proof.

In contrast, manual NER did not demonstrate the same limitations. Because it is performed by a critical human reader capable of distinguishing nuanced linguistic characteristics, the results more closely reflect the texts and its narratives.

Overall, the comparison demonstrates that spaCy's pretrained NER model does not perform well in contexts that deviate syntactically. Ambiguities arising from fictional names that overlap with common nouns reveal the model's reliance on patterns learned from general-domain training data. In contrast, manual annotation, assisted by contextual understanding and narrative awareness, provides more consistent and contextually accurate entity recognition.